

AI TRAINING – WEEK 2

ASSIGNMENT 1

Lists, Tuples & Sets

Learn → Practice → Build → Explain

1. What You Will Learn

You will learn three important Python collection types: lists, tuples, and sets. You will practice accessing and modifying them, identify common errors, and build two small real-world utilities.

- Lists – indexing, slicing, methods, and mutability
- Tuples – indexing, slicing, methods, and immutability
- Sets – unique values and common set operations
- Difference between list, tuple, and set
- Common errors and how to fix them
- Build a simple shopping cart
- Remove duplicate values using a set

2. Student Deliverable – Keep It Simple

Submit ONE Jupyter Notebook only.

File name: Week2_Day1_Lists_Tuples_Sets_YourName.ipynb

Your notebook should contain:

- Part A – Basic practice
- Part B – Lists
- Part C – Tuples
- Part D – Sets
- Part E – Debugging
- Part F – Shopping Cart mini project
- Part G – Duplicate Removal utility
- Part H – Explain what you learned

No separate Python files, Word/PDF files, or screenshots are required.

3. Part A – Understand the Basics

Create and print:

```
fruits = ["apple", "banana", "orange"]
coordinates = (10, 20)
numbers = {1, 2, 3, 4}
```

- What is a list?
Ans: list is built in data structure .it is used to store values of different data types.it is mutable.
- What is a tuple?

Ans:tuple is collection of variables .it stores different data types .it is immutable.once the value is inserted it cannot be changed again.

- What is a set?

Ans:set is a collection of unique values .it avoid to store the same values again and again.

- Which one allows duplicate values?

Ans:yes tuple and list allows duplicate values.but dictionary allows duplicate values not duplicate keys.set never allows duplicate values.

- Which one is mutable?

Ans:list ,set,dictionary are mutable.

4. Part B – List Practice

```
fruits = ["apple", "banana", "orange", "mango", "grapes"]
```

- Print the first item.

Ans:print(fruits[0]))

- Print the last item.

Ans:print(fruits[-1]))

- Print the first three items using slicing.

Ans:print(fruits[1:4]))

- Change "banana" to "watermelon".

ans:fruits[1]='watermelon'

- Add "kiwi".fruits[1]='watermelon'

- Remove "orange".

Ans:fruits.remove('orange')

- Print the final list.

Ans:Print(fruits)

- Find the length of the list.

Ans:print(len(fruits))

Practice: append(), insert(), remove(), pop(), sort(), reverse(), len()

5. Part C – Tuple Practice

```
student = ("Phani", 34, "Hyderabad", "Python")
```

- Print the first item.

```
student = ("Phani", 34, "Hyderabad", "Python")
print(student[0])
```

- Print the last item.

```
student = ("Phani", 34, "Hyderabad", "Python")
print(student[-1])
```

- Print the first two items using slicing.

```
student = ("Phani", 34, "Hyderabad", "Python")
print(student[0:2])
```

- Find the length.

```
student = ("Phani", 34, "Hyderabad", "Python")
print(len(student))
```

- Try changing 34 to 35.
Ans:It cannot change because tuple is immutable.
- Observe the error and explain why it happened.
Ans:it does not support item assignment after creation.

Remember: tuples are generally used when the collection should not be changed.

6. Part D – Set Practice

```
numbers = {10, 20, 30, 20, 10, 40}
```

- Print the set.

```
numbers = {10, 20, 30, 20, 10, 40}
print(numbers)
```

- Observe what happened to duplicates.
Ans:it removes the duplicate elements.
- Add 50.

```
numbers = {10, 20, 30, 20, 10, 40}
numbers.add(50)
print(numbers)
```

- Remove 20.

```
numbers = {10, 20, 30, 20, 10, 40}
numbers.remove(20)
print(numbers)
```

- Check whether 30 exists.

```

numbers = {10, 20, 30, 20, 10, 40}
if 30 in numbers:
    print("yes")
else:
    print("no")

```

- Find the number of unique values.

```

numbers = {10, 20, 30, 20, 10, 40}
print(len(numbers))

```

ans: it will return the length of unique values because set does not allow duplicates.

Important: sets are unordered collections. Do not depend on a particular display order.

7. Part E – List vs Tuple vs Set

Feature	List	Tuple	Set
Ordered?	yes	yes	no
Allows duplicates?	yes	yes	no
Mutable?	yes	no	yes
Supports indexing?	yes	yes	no
Main use?	To store values which may need changes.	To avoid changes	To avoid duplicate values

8. Part F – Quick Debugging

Find the error, fix it, and explain the mistake.

Debugging Problem 1

```

fruits = ["apple", "banana", "orange"]
print(fruits[5])

```

ans: there are only 3 elements in that list but we are asking for 5th element.

```

fruits = ["apple", "banana", "orange"]
print(fruits[2])

```

Debugging Problem 2

```

numbers = (10, 20, 30)
numbers[0] = 100

```

ans: tuple does not add new values once it is created.

Debugging Problem 3

```

items = {1, 2, 3}
print(items[0])

```

ans: set does not allow indexing.

Debugging Problem 4

```
fruits = ["apple", "banana"]
fruits.add("orange")
```

ans: list does not have add() method instead of that it uses append()

9. Part G – Hands-on Activity 1: Shopping Cart

Start with: `cart = ["Rice", "Milk", "Bread"]`

Build a simple shopping cart using a list. Your program should:

- Display the cart.
- Add a new item.
- Remove an item.
- Display the number of items.
- Check whether a particular item exists.
- Display the final cart.

Bonus: create `add_item()`, `remove_item()`, and `show_cart()` functions.

```
cart = ["Rice", "Milk", "Bread"]
print(cart)
cart.append("Eggs")
print(cart)
cart.remove("Milk")
print(cart)
print("Number of items:", len(cart))
if "Bread" in cart:
    print("Bread is available in the cart")
else:
    print("Bread is not available in the cart")
print("Final cart:", cart)
```

10. Part H – Hands-on Activity 2: Duplicate Removal

Create a program that removes duplicate values from a list using a set.

Example input: `[10, 20, 10, 30, 20, 40, 30]`

Expected result: a collection containing only unique values.

```
lst=[10,20,10,30,20,40,50]
print(list(set(lst)))
```

output:

`[40, 10, 50, 20, 30]`

Test with:

- `[1, 1, 2, 2, 3, 3]`

```
lst=[1, 1, 2, 2, 3, 3]
print(list(set(lst)))
```

output:

[1, 2, 3]

- ["apple", "banana", "apple", "orange", "banana"]

```
lst=["apple", "banana", "apple", "orange", "banana"]
print(list(set(lst)))
```

output:

['banana', 'apple', 'orange']

- [10, 20, 30]

Ans:

Remains same because all are unique.

Think: Why is a set useful for this problem?

11. Part I – Explain in Your Own Words

1. What is the main difference between a list and a tuple?

Answer: list can store duplicate values but tuple cannot store duplicate values.

2. Why can't we normally change an item inside a tuple?

Answer: once a tuple is created, its items cannot be changed, added, or removed.

3. Why are sets useful for removing duplicates?

Answer: because set does not allow duplicates.

4. What does indexing mean?

Answer: indexing means accessing a particular value by its index number.

5. What does slicing mean?

Answer: extract the required part from the sequence.

6. What does mutable mean?

Answer: once it is created we can change the values.

7. When would you choose a list instead of a tuple?

Answer: when I want to make changes to values or remove values.

12. Student Submission Checklist

- I completed all practice sections.
- I tested my code.
- I fixed the debugging problems.
- I completed the Shopping Cart activity.
- I completed the Duplicate Removal activity.

- I explained the concepts in my own words.
- My notebook runs without major errors.
- My filename is correct.

13. After This Assignment, You Should Be Able To

- Create and use lists, tuples, and sets.
- Access values using indexes.
- Use slicing.
- Use common collection methods.
- Understand mutability and immutability.
- Remove duplicates using a set.
- Choose the right collection for a simple problem.

Key Principle: Don't just memorize syntax. Understand when and why to use each collection.